

MATRIX ACADEMYwe enhance intelligence

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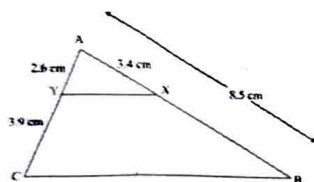
X TEST → QUADRATIC EQUATIONS AND TRIANGLES

Time Allowed: 1 hour

Maximum Marks: 28

Section A

- 1 The equation $x^2 - 8x + k = 0$ has real and distinct roots if [1]
A) $k \geq 8$ B) $k < 16$ C) $k \leq 16$ D) $k > 16$
- 2 The least positive value of k , for which the quadratic equation $2x^2 + kx - 4 = 0$ has rational roots, is [1]
A) $\sqrt{2}$ B) 2 C) ± 2 D) $\pm 2\sqrt{2}$
- 3 In the figure, X and Y are two points on the sides AB and AC respectively in $\triangle ABC$, such that $AX = 3.4$ cm, $AB = 8.5$ cm, $AY = 2.6$ cm and $YC = 3.9$ cm. Which of the following relation is correct? [1]



- A) BC is not parallel to XY B) $3BC = 2XY$ C) $BC \parallel XY$ D) $BC = 2XY$
- 4 If a tree casts a 18 feet shadow and at the same time, a child of height 3 feet casts a 2 feet shadow, then the height of the tree is [1]
A) 45 feet B) 27 feet C) 36 feet D) 32 feet
- 5 Assertion (A): If $5 + \sqrt{7}$ is a root of a quadratic equation with rational co-efficients, then its other root is $5 - \sqrt{7}$. [1]

Reason (R): Surd roots of a quadratic equation with rational co-efficients occur in conjugate pairs.

- A) Both A and R are true and R is the correct explanation of A.
B) Both A and R are true but R is not the correct explanation of A.
C) A is true but R is false.
D) A is false but R is true.

Section B

- 6 The length of a rectangular field is three times its breadth. If the area of the field be 147 sq metres, find the length of the field. [2]
- 7 E is a point on the side AD produced of a parallelogram ABCD and BE intersects CD at F. Show that $\triangle ABE \sim \triangle CFB$. [2]
- 8 In $\triangle ABC$, D and E are points on the sides AB and AC respectively such that $DE \parallel BC$. If $AD = 2$ cm, $AB = 6$ cm and $AC = 9$ cm, find AE. [2]

Section C

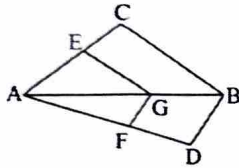
- 9 A dealer sells an article for ₹ 75 and gains as much per cent as the cost price of the article. Find the cost price of the article. [3]

Section D

- 10 A train takes 2 hours less for a journey of 300 km if its speed is increased by 5 km/hr from its usual speed. Find the usual speed of the train. [5]
- 11 If a line is drawn parallel to one side of a triangle intersecting the other sides in distinct points, prove that it divides the other sides in the same ratio. [5]

Use the above result to prove the following:

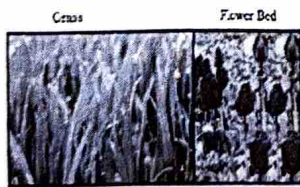
In the given figure, if $EG \parallel CB$ and $FG \parallel DB$, then prove that $\frac{AE}{EC} = \frac{AF}{FD}$.



Section E

- 12 Read the following text carefully and answer the questions that follow:

Shreya has a field with a flowerbed and grassland. The grassland is in the shape of rectangle while flowerbed is in the shape of square. The length of the grassland is found to be 3 m more than twice the length of the flowerbed. Total area of the whole land is 1260 m^2 .



- If the length of the square is x m, then find the total length of the field. (1)
- What will be the perimeter of the whole figure in terms of x ? (1)
- Find the value of x if the area of total field is 1260 m^2 . (2)

OR

- Find area of grassland and the flowerbed separately. (2)